EASWARI ENGINEERING COLLEGE
RAMAPURAM, CHENNAI
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AUTONOMOUS

END SEMESTER EXAM – NOV/DEC 2026

Object Oriented Programming Using Java – 2321CSC304R

EASWARI ERP PORTAL

Intelligent Attendance Control and Student Progress Monitoring Platform

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II Year | B.E. Computer Science and Engineering | E Section

Introduction

- Attendance and internal marks decide exam eligibility (75% rule) and final grades.
- Today they live in paper registers and spreadsheets updated by different people at different times.
- The Easwari ERP Portal puts attendance, marks, results, fees, leave, services and placement into one web platform.

Three portals, one sign-in
Student · Faculty · Administrator

Core idea
Store each fact once and let the server decide who may change it.

Problem Statement

- **No single record –** the same student shows different attendance on different lists.
- **No control over authority –** any teacher can edit a class; corrections overwrite and leave no trail.
- **Late warning –** short attendance is noticed only when the eligibility list is displayed.
- **Scattered services –** fees, leave and placement information sit at different counters.

Objectives

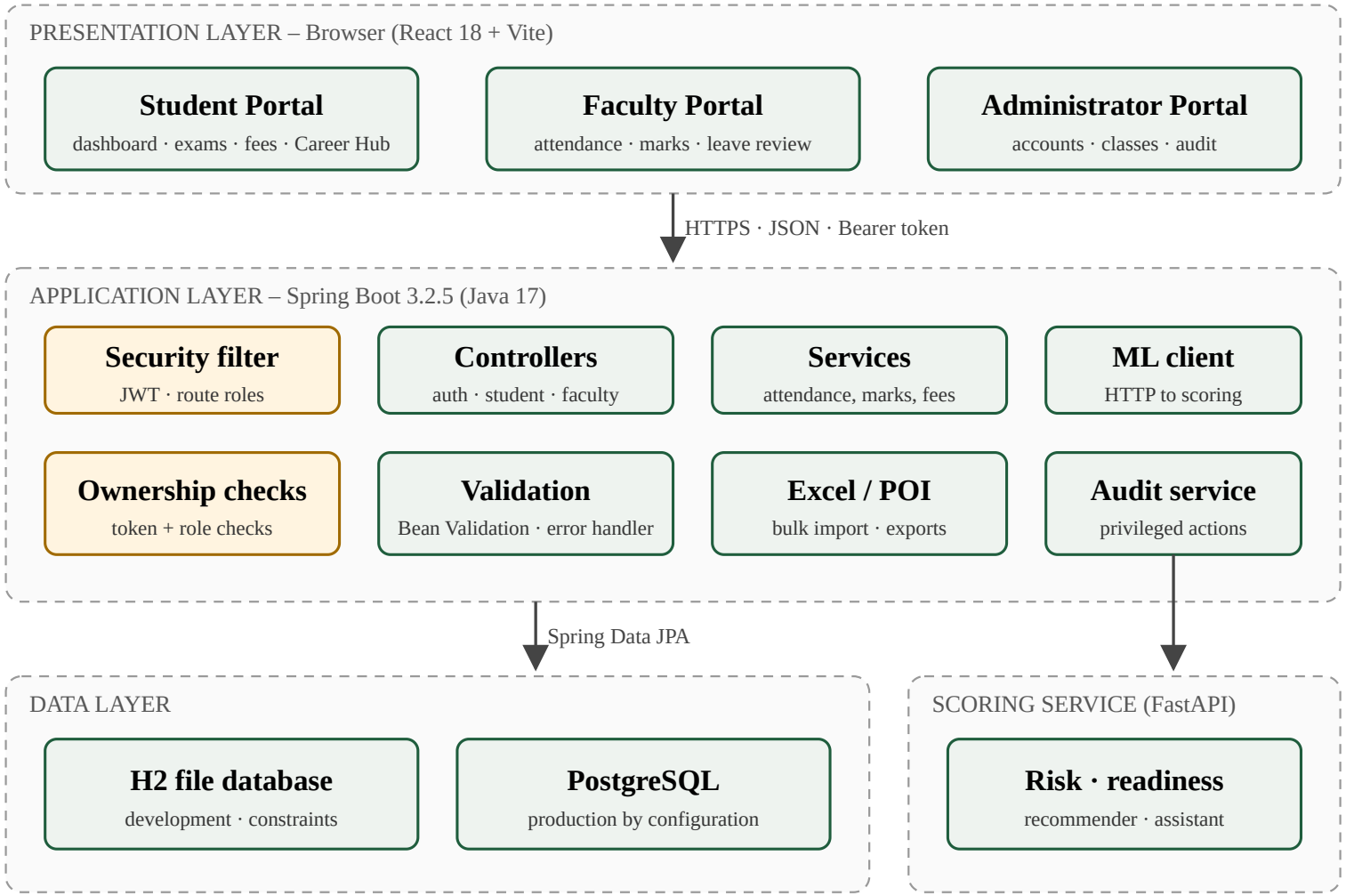
- Student, Faculty and Administrator portals behind one secure sign-in with temporary passwords.
- Period-level attendance with a database constraint that makes double marking impossible.
- Coordinator marks the whole class; subject teacher only his or her own subject.
- Internal marks released to students only after publication.
- Fees, leave / on-duty, campus services, placement and a Career Hub.
- Explainable academic-risk score from a separate scoring service.

Literature Survey

Study	What it does	Gap for our project
Ramadhani & Satria (2025) – QR attendance	QR scan saves attendance to a central database	No rules on who may change a record
Akçapınar et al. (2019) – early warning	Predicts at-risk students from online-learning logs	Needs an LMS; single course
Howard et al. (2017) – prediction methods	Compares 8 methods; continuous assessment is the best predictor	Needs LMS data; supports weighting internal marks

> **Observation:** existing work captures data or predicts risk, but does not define authority over records for an affiliated engineering college.

Proposed System & Architecture

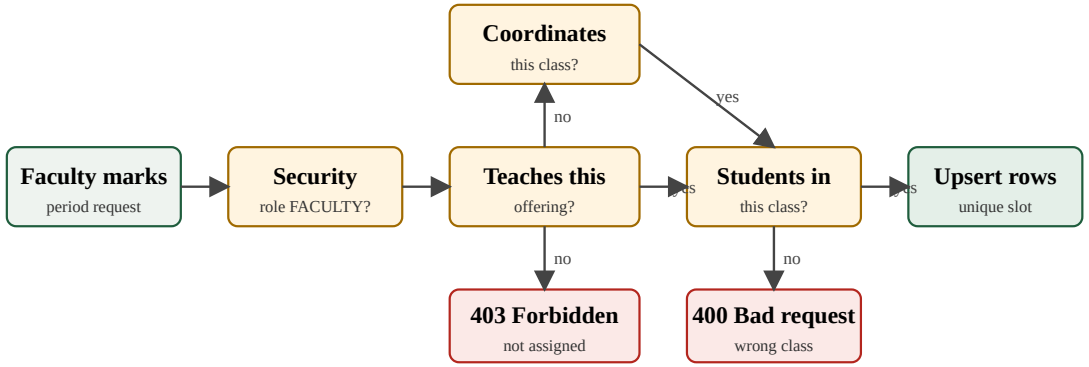


Technology Stack & Modules

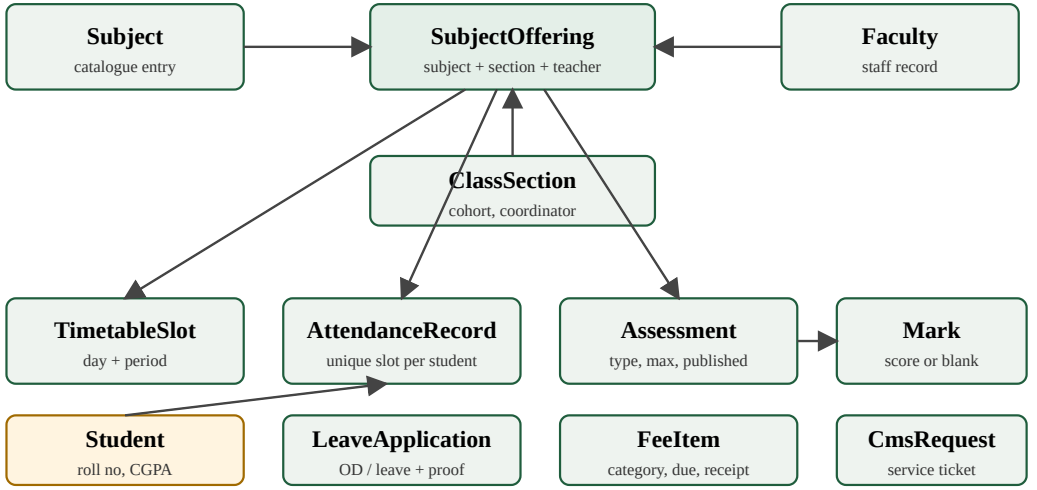
Layer	Technology
Back end	Java 17, Spring Boot 3.2.5, JPA
Database	H2 file DB → PostgreSQL
Security	Spring Security, JWT, BCrypt
Front end	React 18, Vite 5, Recharts
Scoring	Python FastAPI, NumPy
Testing	JUnit 5 (11 tests)

- Authentication & accounts
- Attendance
- Marks & examinations
- Fees, services & leave
- Administration & audit trail
- Risk score & Career Hub

Attendance Authority & Data Model



Unique slot uk_attendance_slot on student + offering + date + period



Security & Sign-in

- Role picker → username + password → signed JWT.
- Temporary password: 10 random characters, forced change at first login.
- Students can set a 6-digit quick code; locked 15 min after 5 wrong tries.
- Token kept in memory only – nothing in local storage.

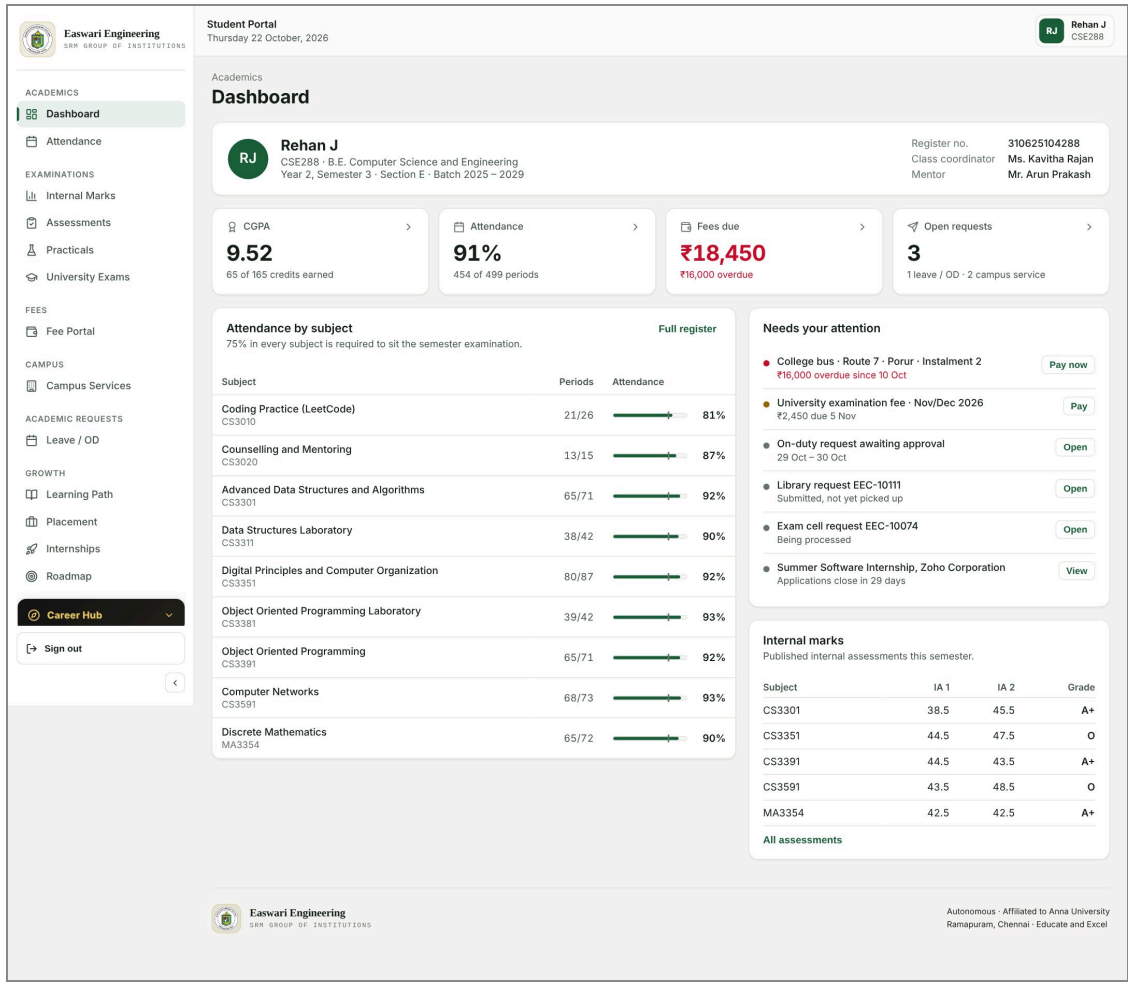
Three layers

Route guards (/api/admin, /api/faculty, /api/student) → role annotations → ownership checks in services.

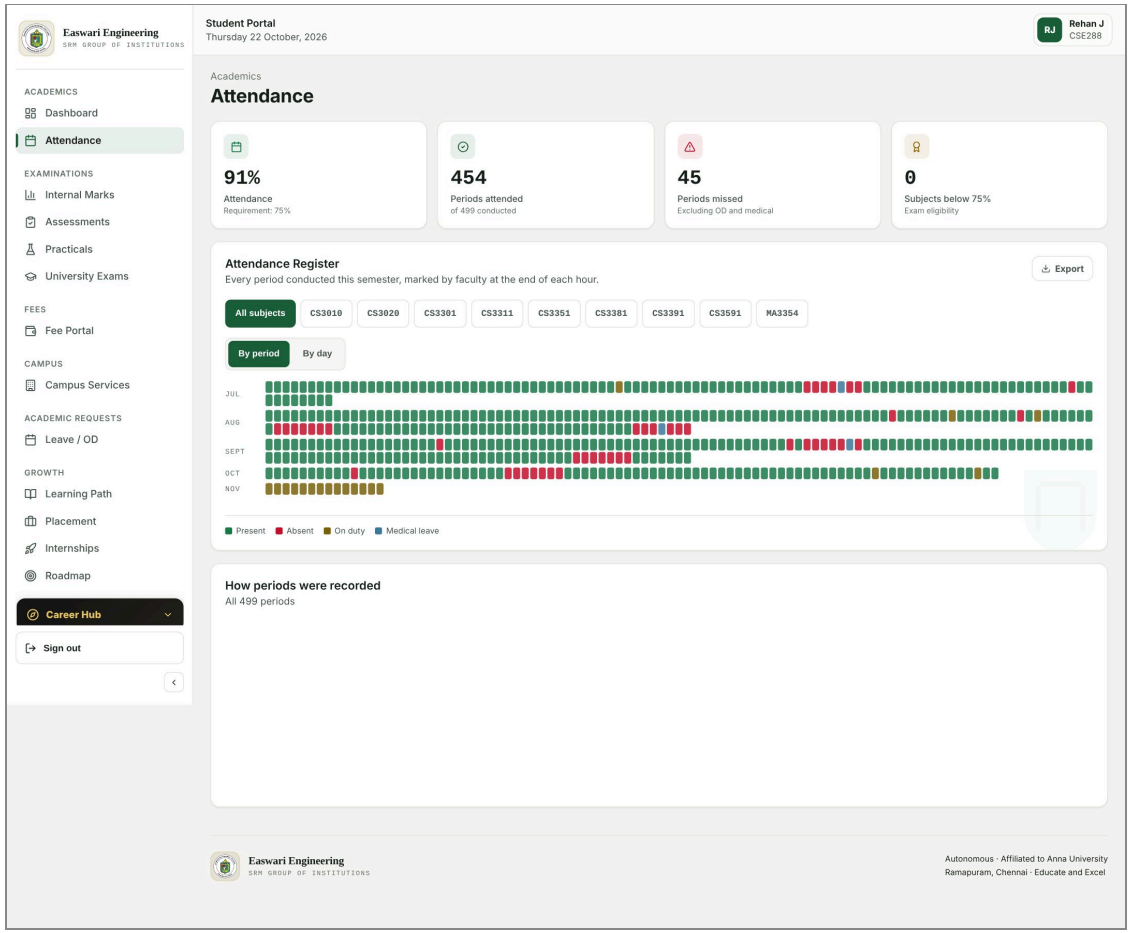
Audit trail

Every privileged admin action is recorded.

Student Portal

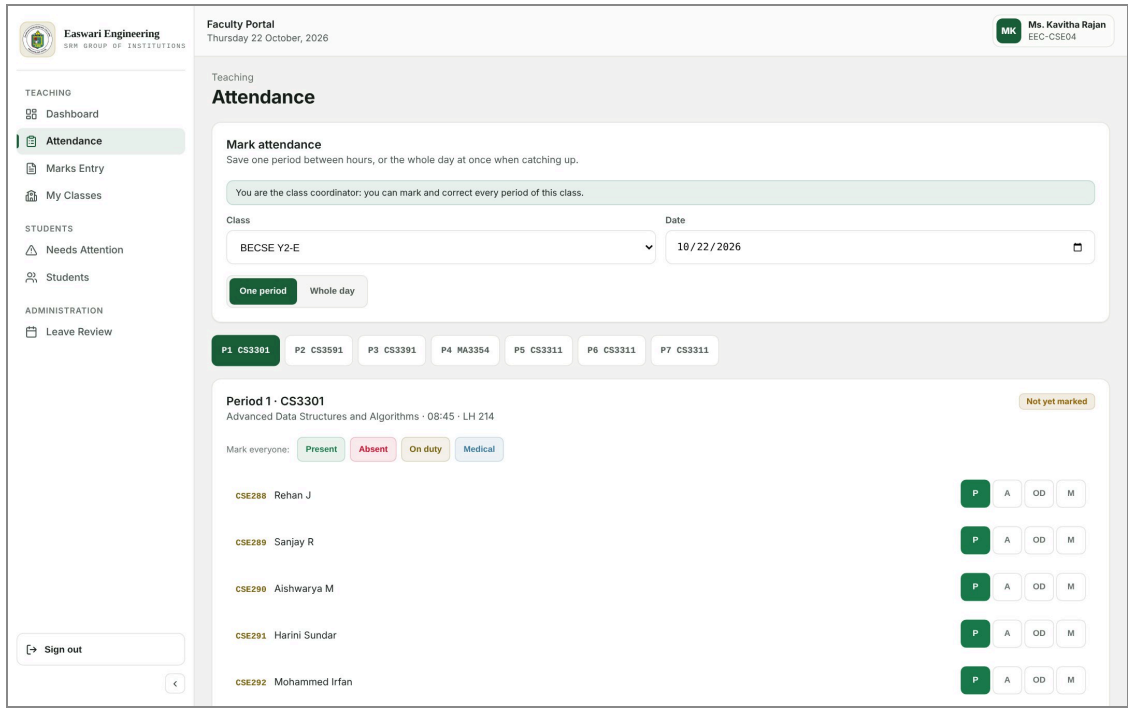


Dashboard

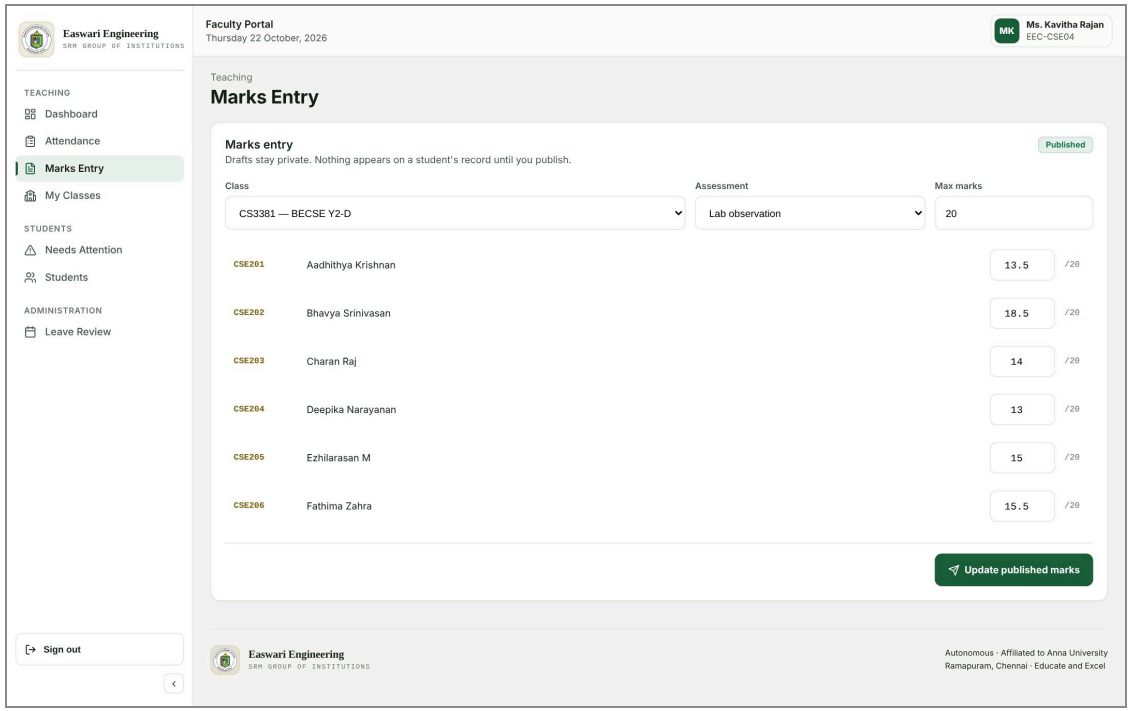


Attendance register

Faculty Portal

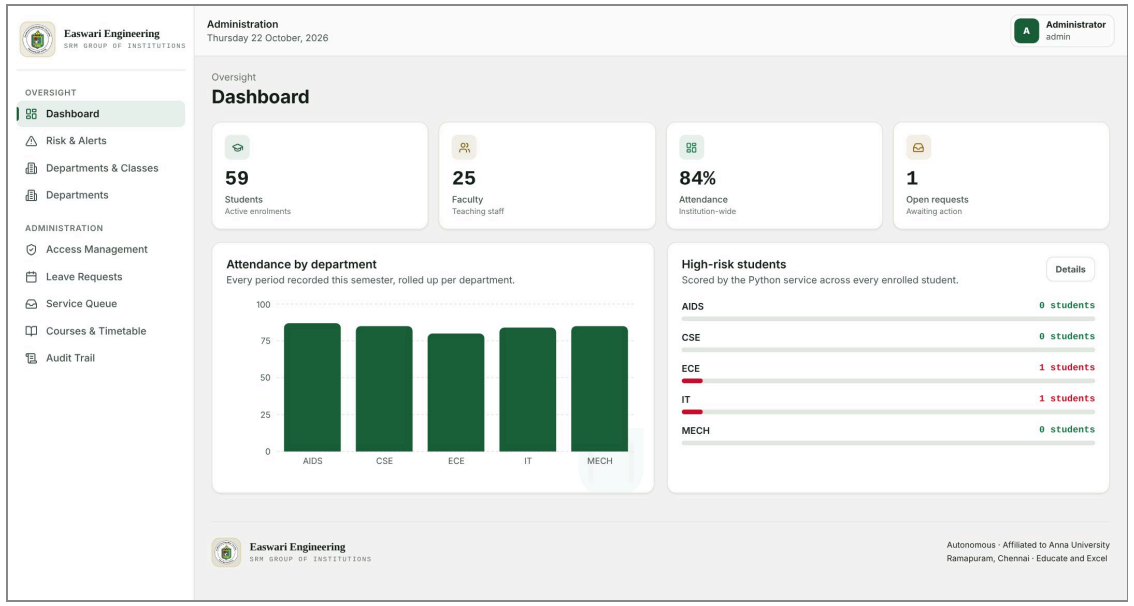


Coordinator attendance entry

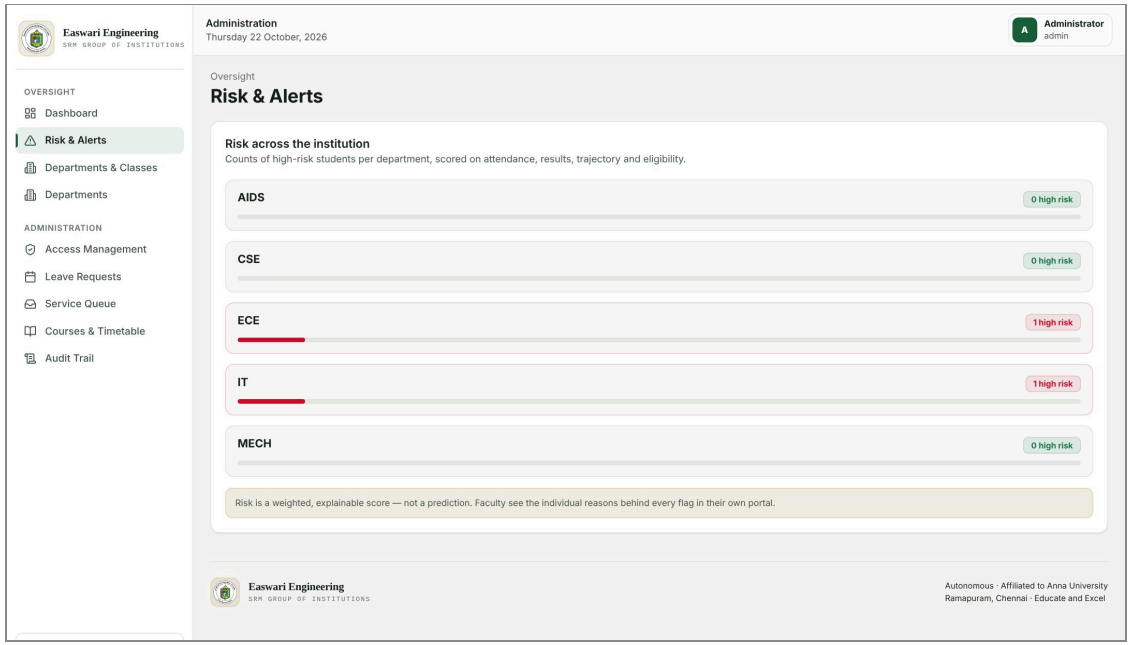


Marks entry

Administrator Portal & Career Hub



Admin dashboard



Risk and alerts

Testing & Results

68
REST test cases

67
passed

11/11
unit tests passed

28,497
attendance records seeded

Profile	Attend.	CGPA	Score	Band
Healthy	93%	8.90	0	LOW
Borderline	78%	7.20	36	MEDIUM
Struggling	71%	6.40	66	HIGH

- Subject teacher refused (403) outside own subject.
- Re-marking a period updates the row – percentage unchanged.
- One defect found: unknown status value gives HTTP 500.

Conclusion, Limitations & Future Scope

Conclusion

Each fact is stored once and each rule is enforced on the server, giving students, faculty and administrators one trustworthy view.

Limitations

Simulated payment · seeded trial data · defects D1 (500 error) and D2 (fixed reset password) · risk weights need calibration.

- Real payment gateway
- Import real college data
- Alerts near 75% attendance
- Rate limiting & two-factor login
- QR / mobile attendance capture
- Parent, mentor, hostel, library modules

Thank You

Questions are welcome

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